

WDI - Statistics for High Dimensional Data

Alessandro Ceschel (0001178799), Maria Gnoli (0001179817),
Matteo Severi (0001174713), Gianluca Antonio Spennacchio (0001183602)

June 26, 2025

1 Introduction

The WDI dataset (the primary World Bank collection of development indicators) contains observations for 100 countries and 32 regions¹ in the world, comprising 8 continuous indicators (expressed in % of GPI) measured in 1993: **Gross**, **Carbon**, **Consume**, **Education**, **Energy**, **Mineral**, **Forest**, **Particulate** as described in Table 1. They represent both economically-grounded indicators - focusing on aspects like gross savings and capital consumption - and environmental-damage measures.

Variable	Description
Gross	Difference between gross national income and public and private consumption, plus net current transfers.
Carbon	Cost of damage due to carbon dioxide emissions times the number of tons of CO ₂ emitted.
Consume	Consumption of fixed capital (replacement value of capital used up in the process of production).
Education	Current operating expenditures in education, including wages and salaries and excluding capital investments in buildings and equipment.
Energy	Energy depletion (ratio of the value of the stock of energy resources to the remaining reserve lifetime – capped at 25 years).
Mineral	Mineral depletion (ratio of the value of the stock of mineral resources to the remaining reserve lifetime – capped at 25 years).
Forest	Net forest depletion (product of unit resource rents and the excess of roundwood harvest over natural growth).
Particulate	Particulate emission damage, due to exposure of a country’s population to ambient concentrations of particulates measuring less than 2.5 microns in diameter (PM2.5) (calculated as foregone labor income due to premature death).

Table 1: Descriptions of the selected sustainability indicators

The overall objective of this report is to grasp the underlying structure of the data based on the environmental and economic sustainability indicators, identifying groups and verifying their robustness.

Our analysis is structured in three main steps. First, we apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while preserving as much of the original variability as possible. We then perform a cluster analysis on the same data and project the results on the principal components, aiming to identify groups of countries with similar development profiles - potentially independent of their geographical location or position in the global income distribution. Lastly, we conduct a Discriminant Analysis to evaluate whether the identified clusters reflect an underlying structure in the data.

2 Methodological Foundations

In this section, we provide a theoretical explanation of the methodologies applied in the project. Each analytical choice will then be justified in its corresponding section.

The first goal of our analysis is to reduce the dimensionality of the dataset, making interpretation of our results easier, without losing critical information contained in the original data. An effective approach to achieve this goal is to use *Principal Component Analysis* (PCA) to find linear combinations of the original variables that best explain the variability of the data.

Formally, PCA transforms a set of d correlated variables $\mathbf{X} = [X_1, \dots, X_d]$ into a set of d uncorrelated variables $\mathbf{W} = [W_1, \dots, W_d]$, known as principal components, so that:

$$\text{Cov}(W_k, W_{k'}) = 0 \quad \forall k \neq k'$$

The variance of such components collectively accounts for the total variance of the dataset:

$$\sum_{k=1}^d \mathbb{V}(W_k) = \sum_{i=1}^d \mathbb{V}(X_i)$$

The components are then sorted in descending order based on the portion of variance explained by each:

$$\mathbb{V}(W_1) > \mathbb{V}(W_2) > \dots > \mathbb{V}(W_d)$$

¹Both geographical and economic (e.g. Latin American and Caribbean; High income)

More precisely, the principal components are computed in the following way. Consider a random vector $\mathbf{X} \sim (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and the spectral decomposition of the covariance matrix, given by

$$\boldsymbol{\Sigma} = \boldsymbol{\Gamma} \boldsymbol{\Lambda} \boldsymbol{\Gamma}',$$

where $\boldsymbol{\Lambda}$ is a diagonal matrix containing the eigenvalues $\lambda_1 > \lambda_2 > \dots > \lambda_d > 0$, and $\boldsymbol{\Gamma}$ is an orthogonal matrix whose columns $\boldsymbol{\eta}_k$ are the eigenvectors of $\boldsymbol{\Sigma}$, each corresponding to its respective eigenvalue λ_k .

The eigenvectors in $\boldsymbol{\eta}_k$ contain the loadings of the k -th principal component, representing the “weight” of each observed variable on the principal component. Therefore, the k -th principal component is a scalar given by:

$$W_k = \sum_{i=1}^d \eta_{ki} X_i = \boldsymbol{\eta}_k' \mathbf{X}$$

and its variance corresponds to the eigenvalue λ_k .

To avoid making the variance of any W as large as wanted by increasing the corresponding $\boldsymbol{\eta}_k$, loadings must satisfy the following restrictions:

$$\begin{aligned} \sum_{k=1}^d \eta_{ki}^2 &= 1 \quad (\text{to get a unique solution}) \\ \sum_{k=1}^d \eta_{kj} \eta_{ki} &= 0, \quad j \neq i \quad (\text{to get orthogonal components}) \end{aligned} \tag{1}$$

Finally, it is possible to select the first r principal components from the set of d original components, based on the methods described in Section 4.1 .

We then conduct a **Cluster Analysis** (CA), which goal is to determine the cluster allocation of each observation and the optimal number of clusters.

The first step of cluster analysis is the measurement of the distance (since our variables are continuous) apart of all the possible pairs of objects. This is done computing the *Manhattan distance* to place less weight on large differences in the variables at individual levels and on outliers values. It is defined as: $\delta_{ij} = \sum_{k=1}^d |x_{ik} - x_{jk}|$.

The second step is to apply a proper clustering method. In this project we compared both hierarchical and non-hierarchical methods. Hierarchical methods can be both agglomerative and divisive. Here we focus on agglomerative hierarchical methods in which the starting points are single observations treated as one-member clusters, that are then grouped step-by-step based on the distance between such clusters. The methods used to carry out agglomerative hierarchical methods are nearest neighbor, farthest neighbor, centroid, and Ward method.

In particular, this project employs the *complete linkage method* (or farthest neighbor), which uses the largest distance between members of different clusters, and the *Ward method*, which is based on the concept of least lost information caused from the amalgamation of a pair of clusters. To compute such loss in information the within sum of squares is used: $W_s = \sum_{k=1}^d \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$. Cluster aggregation is based on minimizing such value.

Non-hierarchical methods instead assume the number of clusters k a priori. The most used method is the *k-means clustering*. Partitioning is based on the minimization of the variability within the clusters.

In the final part of our analysis, **Discriminant Analysis** (DA) was applied as a supervised classification technique to assign observations to clusters previously obtained via unsupervised *k*-means clustering, with the aim of assessing the statistical coherence of the clustering solution. We first implemented **Linear Discriminant Analysis** (LDA) and subsequently applied **Quadratic Discriminant Analysis** (QDA) to address potential violations of the homoscedasticity assumption.

LDA assumes that each cluster (class) $\mathcal{C}_k$ follows a multivariate normal distribution with common covariance matrix $\boldsymbol{\Sigma}$ and class-specific mean $\boldsymbol{\mu}_k \in \mathbb{R}^d$:

$$X | \mathcal{C}_k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}), \quad k = 1, \dots, K.$$

The LDA discriminant score is:

$$\delta_k(x) = x^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k + \log \pi_k$$

QDA generalizes this by allowing each class its own covariance matrix $\boldsymbol{\Sigma}_k$, leading to:

$$x | \mathcal{C}_k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \delta_k(x) = -\frac{1}{2} \log \det(\boldsymbol{\Sigma}_k) - \frac{1}{2} (x - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (x - \boldsymbol{\mu}_k) + \log \pi_k.$$

where π_k is the prior probability of class $\mathcal{C}_k$, commonly estimated by the empirical class proportion $\pi_k = \frac{n_k}{n}$. The decision rule r then assigns x to the class with the maximum discriminant score:

$$r(x) = \arg \max_k \delta_k(x).$$

To evaluate performance, we used **Leave-One-Out Cross-Validation** (LOOCV), which provides a nearly unbiased estimate of classification error and is particularly suitable when the sample size n is moderate, as it makes efficient use of all available data.

3 Descriptive Analysis

To carry out our analysis, we chose to focus exclusively on the observations corresponding to individual countries, excluding the regional aggregates (e.g. Euro Area). This decision helps us avoid redundancy, as regional values inherently reflect the data from the countries they encompass. More importantly, it aligns with our objective of identifying groups of countries that share similar development patterns, as revealed by the rich dataset of 100 country-level observations. We started by verifying the absence of NA values and of duplicated rows before conducting the analysis.

A preliminary graphical exploration of the data reveals that most variables are heavily skewed. To address this, we applied a log-transformation to all variables², except for **Consume** and **Gross**. These two variables were already approximately normally distributed, and applying a log-transformation would have unnecessarily compressed their variance, which is valuable for the statistical techniques used in our analysis. To further support our decision, we assessed the univariate normality of **Gross** and **Consume** using the *Shapiro-Wilk test*, which led us to reject the null hypothesis of Normal distribution ($p < 0.05$), both before and after log-transformation. Corresponding histograms and Q-Q plots, provided in the Appendix, illustrate that the log-transformation did not improve normality for these two variables, and in fact slightly worsened it, so we proceeded with the original (non-log) variables in our analysis.

Following the log-transformation, all variables were standardized. The distributions of the log-transformed and standardized variables are presented below in Figure 1.

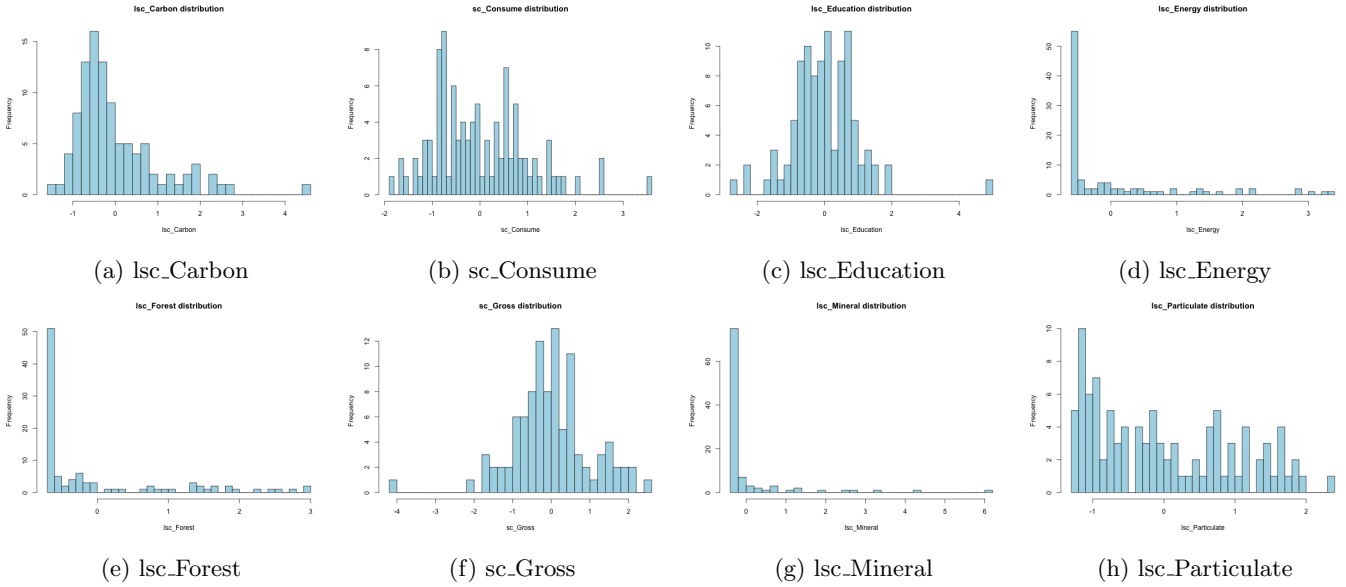


Figure 1: Histograms of selected variables

We continued the explorative analysis by visualizing the correlation matrix and observing the descriptive statistics. Both are presented below in Figure 2 and Table 2.

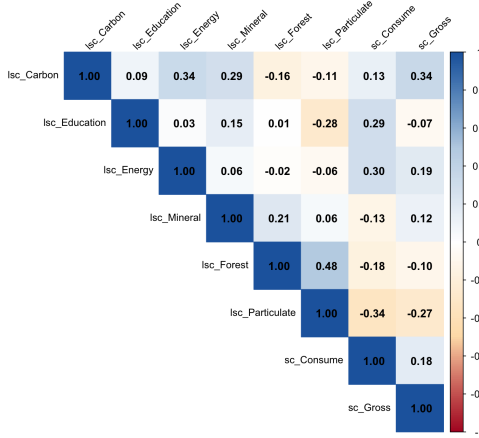


Figure 2: Correlation Matrix for selected variables

Variable	Mean	SD	Min	Max
Carbon	1.7361	2.3608	0.0050	22.4096
Consume	11.0612	5.1080	0.4638	30.6909
Education	4.2226	3.3541	0.4852	37.7216
Energy	2.0084	4.9961	0.0000	30.5366
Gross	19.9163	10.6660	-27.8908	49.5651
Mineral	0.1846	0.5904	0.0000	5.4525
Forest	1.5388	3.5967	0.0000	20.2798
Particulate	1.6197	1.6846	0.0407	8.5327

Table 2: Descriptive statistics for each variable

²Methodological note: since several variables had very small values and the variable Energy included zeros we added 1 to all values in applying the log transformation.

We observe that **Energy**, **Mineral** and **Forest** still exhibit notably skewed distributions that deviate from normality, even after the log transformation. Furthermore, **Particulate** too has a non-normal distribution even after the transformation, but it is more constant than before such refinement. Nevertheless, we choose to retain these variables to avoid excessively reducing the dimensionality of our dataset. On the other hand, due to its very low correlation with all other variables, the variable **Education** is excluded from the analysis³. All subsequent analyses are therefore conducted on a modified dataset consisting of log-transformed and standardized variables, with **Education** removed.

4 Analysis

4.1 Principal Component Analysis (PCA)

As we mentioned, the first step of our analysis involves reducing the dimensionality of the dataset through PCA. Another important objective of this procedure is to simplify interpretations of clustering and discriminant analysis. For this we decided to conduct PCA even if the data do not show high correlations, with the highest being the one between **lsc.Particulate** and **lsc.Forest** equal to 0.48 (Figure 2).

To understand which is the optimal number of principal components to retain, we have to follow some theoretical principles. First of all, the portion of variance explained by the first r principal components must be such that most of the original information is kept. This can be done looking at the scree plot in Figure 3 and at Table 3.

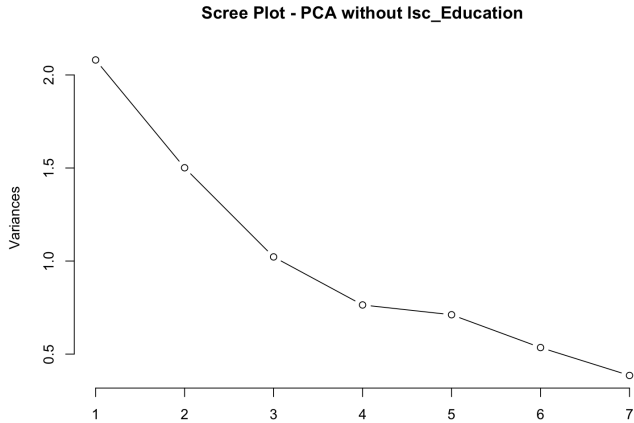


Figure 3: Scree Plot for PCA

PC	SD	% Var.	Cum. %
PC1	1.4424	29.72	29.72
PC2	1.2253	21.45	51.17
PC3	1.0111	14.60	65.77
PC4	0.8741	10.91	76.69
PC5	0.8435	10.16	86.85
PC6	0.7315	7.64	94.50
PC7	0.6206	5.50	100.00

Table 3: Standard Deviation and Explained Variance

The graph suggests to retain 3 or 4 principal components (in correspondence of the "elbow" in the graph). While, from the table it can be noted how no single component overwhelmingly dominates the variance structure; indeed, the first three principal components together account for approximately 66% of the total variance and provide distinct, interpretable insights into the underlying data-generating processes. A possible explanation for this moderate value may be linked to the correlation matrix problem already described above.

The second principle to be met is to select components with eigenvalues greater than 1, following the *Kaiser criterion*. In our analysis, it would imply retaining the first three principal components, with eigenvalues equal to $\lambda_1 = 2.081$, $\lambda_2 = 1.501$, and $\lambda_3 = 1.022$.

Furthermore, prioritizing components that offer meaningful and interpretable insights is the logical reading key to our data reduction.

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7
lsc_Carbon	-0.583	0.536	-0.133	0.359	0.140	-0.376	0.253
lsc_Energy	-0.504	0.360	0.602	0.259	0.055	0.428	-0.035
lsc_Mineral	-0.033	0.726	-0.386	-0.038	-0.522	0.100	-0.194
lsc_Forest	0.549	0.523	0.329	-0.444	-0.042	-0.001	0.343
lsc_Part particulate	0.675	0.410	0.297	0.152	0.265	-0.268	-0.351
sc_Consume	-0.607	-0.197	0.497	-0.277	-0.356	-0.354	-0.128
sc_Gross	-0.600	0.275	-0.222	-0.519	0.466	0.057	-0.157

Table 4: Principal Component Loadings for Each Variable

³Besides public education expenditures being considered an addition to savings, because of the wide variability in their effectiveness, this cannot represent the value of investments in human capital. The calculation should also consider private education expenditure, but data are not available for many countries.

The **first principal component** can be interpreted as a Sustainability Trade-off Gradient, which captures the major axis of variability across countries - as it accounts for the 27% of the total variance. It differentiates countries according to their position on a continuum of environmental and institutional vulnerability versus resource-intensive economic activity.

On the positive end of the gradient, countries exhibit high **Particulate** emission damage (loading = +0.675) and substantial net **Forest** depletion (+0.549). These indicators signal local environmental harm, affecting air quality and terrestrial ecosystems. Such patterns are often associated with weak regulatory institutions, high dependence on biomass and land exploitation, and limited investment capacity for mitigation or environmental protection. This is confirmed by low gross savings (−0.600), suggesting limited future-oriented economic planning or reinvestment. This framework reflects structural fragility in both ecological and economic dimensions.

Conversely, countries on the negative end are characterized by high **Gross** savings, high consumption of fixed capital (**Consume** = −0.607), and significant **Carbon** (−0.583) and **Energy** depletion (−0.504). Countries at this end are pursuing a capital-intensive, fossil-fuel dependent model of growth, which may be economically efficient but ecologically destabilizing. Indeed, they invest heavily in their capital base and exhibit active reinvestment dynamics. However, these gains come at the cost of extensive fossil fuel use and long-term environmental pressure, particularly in terms of non-renewable resource depletion.

The **second component** can again be thought of as a gradient, this time dominated by a strong positive loading on **Mineral** (0.726), with additional substantial contributions from **Carbon** (0.536), **Forest** (0.523), and **Particulate** (0.410). This homogeneously positive loading structure suggests that higher PC2 values are associated to countries experiencing simultaneous environmental stress across several dimensions. Economically, it may correspond to countries heavily reliant on natural resource extraction and subject to both global (carbon emissions) and local (particulate) externalities.

The **third component** features strong positive loadings on **Energy** (0.602) and **Consume** (0.497), contrasted with a negative loading on **Mineral** (−0.386). It captures a dividing pattern between countries with intensive domestic use of energy and capital - potentially industrializing or industrialized economies (at the higher end of PC3 continuum), and those primarily engaged in mineral extraction with relatively lower capital consumption (at the lower end). The latter may reflect resource-exporting nations whose extractive activities are oriented outwards, with limited domestic productive transformation.

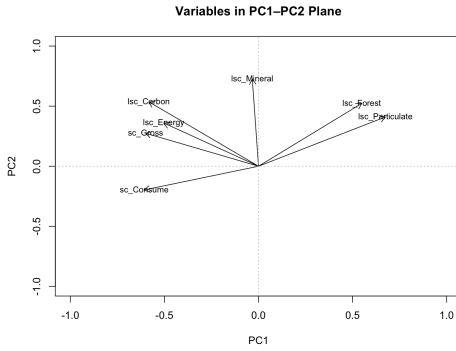


Figure 4: Loading plot PC1 vs PC2

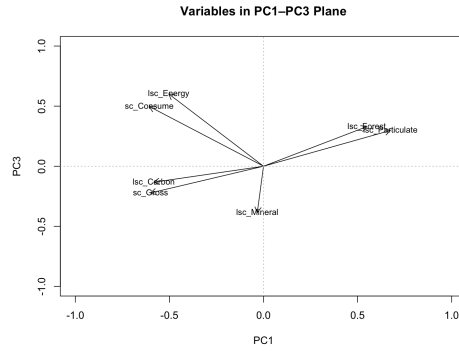


Figure 5: Loading plot PC1 vs PC3

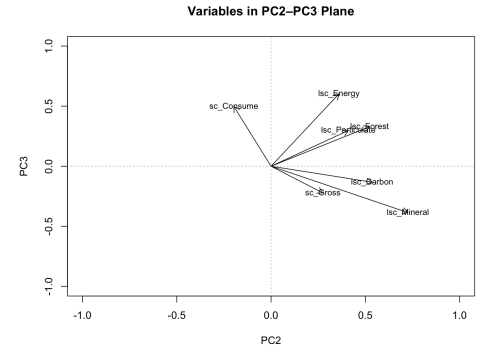


Figure 6: Loading plot PC2 vs PC3

The biplots corresponding to the first three principal component planes (PC1–PC2, PC1–PC3, and PC2–PC3) visually summarize how variables - that appears as arrows (loadings) - relate in the reduced PCA space.

The direction of a vector shows how a variable aligns with the components, while its length reflects the variable's contribution to the PC space. Importantly, the angle between vectors reveals correlation: small angles imply positive correlation, right angles indicate no correlation, and angles near 180° suggest negative correlation.

In Figure 4, **lsc.Particulate** and **lsc.Forest** are closely aligned and load positively on PC1, suggesting that countries with high particulate emissions also experience forest depletion. On the opposite side of PC1, we observe **sc.Gross**, **sc.Consume**, **lsc.Carbon**, and **lsc.Energy**, indicating that countries with higher savings and capital depreciation also show higher carbon and energy depletion. This supports the trade-off interpretation previously discussed, where capital-intensive growth patterns contrast with visible ecological degradation.

In Figure 5, **lsc.Energy** and **sc.Consume** again load strongly and positively on PC3, reflecting a pattern of domestic energy use and capital turnover. These variables are nearly orthogonal to **lsc.Mineral**, which loads negatively on PC3 and is oriented differently, suggesting that countries with mineral depletion tend to differ in structure from those with energy-intensive consumption.

Finally, Figure 6 confirms the unique position of **lsc.Mineral**, which is strongly aligned with PC2 and lies nearly orthogonal to **lsc.Energy**. The general pattern of positive loadings of PC2 is confirmed; the directions of **lsc.Mineral** and **lsc.Consume** also corroborate the interpretation of PC3 which comprehends countries with high mineral depletion mainly for export.

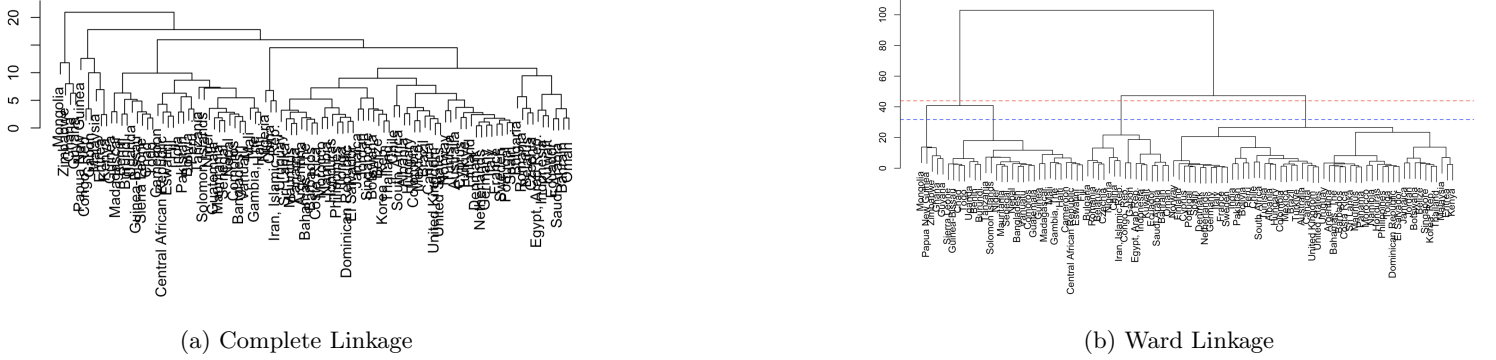
Overall, the biplots confirm the internal coherence of the PCA: they validate the major trade-offs and sustainability dimensions observed quantitatively, offering a compact visual synthesis for clustering and classification.

To sum up, the previous analysis led us to choose three principal components following the *Kaiser criterion* and for interpretation purposes. This structure provides an empirical foundation for subsequent clustering and classification analyses, facilitating the identification of meaningful country groupings, not just by geography or income, but also based on environmental and economic sustainability profiles.

4.2 Cluster Analysis

In this section we perform a cluster analysis with the goal of analyzing the different types of countries and potential country groupings. After having conducted a PCA to comprehend the main drivers of the dataset's variability, we utilize the same data as in the exploratory analysis to conduct the clustering, so the results can subsequently be projected onto the principal component dimensions.

Hierarchical Clustering We first explored hierarchical clustering using Manhattan distance to construct the distance matrix, as this method gives less weight to outlier values. Although we initially computed dendrograms using single linkage and centroid methods, we chose not to include them since they produced unbalanced clusters without meaningful interpretations. Instead, in the following figure, we present the dendrograms using complete linkage and Ward methods.



In the complete linkage method we notice the presence of nesting, which creates unbalanced clustering at whatever height one chooses to cut. For this reason, we continue the analysis with the Ward method.

Visual inspection of the Ward method dendrogram reveals potentially balanced clusters when cutting at either three or four clusters (shown by the red and blue lines in the figure above). To corroborate this, we will now conduct a non-hierarchical cluster analysis.

Non-Hierarchical Clustering In this section, we use another clustering method that requires specifying the number of clusters in advance. Based on our previous analysis, we hypothesized that four clusters would be optimal. To verify this, we examined the scree plot. As Figure 8 shows, increasing the number of clusters beyond four provides diminishing returns in terms of within and between sum of squares. Using the elbow method, the distinction between three and four clusters is subtle, as the curve flattens between these points.

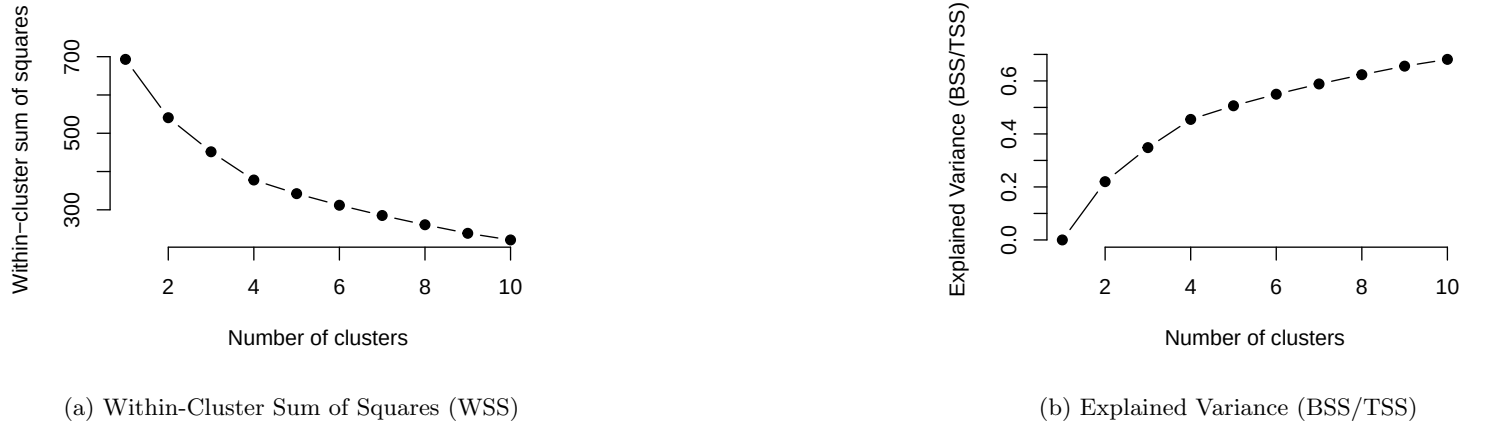
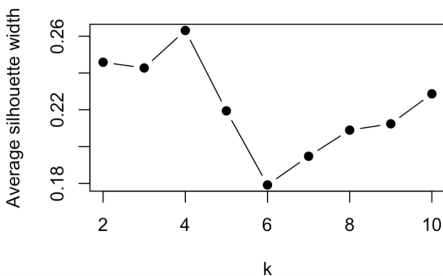


Figure 8: Diagnostics plots to determine the number of clusters

Figure 9: Silhouette index plot



To determine the optimal number of clusters more precisely, we used the Silhouette index, a measure that compares how similar an object is to its own cluster versus other clusters. Figure 9 shows the average Silhouette index values for the k -means method. After considering the combined evidence from the Silhouette index, WSS, and BSS, we conclude that $k = 4$ is the best choice.

We see that the k -means method suggests four clusters. The first cluster is composed of 48 countries, the second has 33 members, the third 15, and the fourth 4.

To understand the differences between the clusters, we now proceed with a descriptive analysis of the variables that most characterize the groups found using k -means.

	lsc_Carbon	lsc_Energy	lsc_Mineral	lsc_Forest	lsc_Part particulate	sc_Consume	sc_Gross
Cluster 1	-0.0954	-0.3231	-0.1172	-0.5417	-0.7580	0.2386	0.2102
Cluster 2	-0.4547	-0.4419	-0.1969	0.6588	1.1439	-0.5050	-0.6379
Cluster 3	0.9628	1.9661	-0.2798	-0.0873	-0.1835	0.5576	0.5742
Cluster 4	1.2851	0.1501	4.0799	1.3924	0.3469	-0.7880	0.5863

Table 5: Means by Cluster (k -means method)

Cluster 1 – Balanced, moderate economies (n = 48) This cluster is composed of countries exhibiting relatively low to moderate levels of environmental impact across all indicators. Carbon and energy depletion, mineral use, and forest loss are all near zero, while particulate pollution is notably low. On the economic side, these countries show modest levels of capital consumption and moderate gross savings. Overall, they likely represent economies with balanced environmental and financial profiles - possibly middle - to high-income countries with a strong emphasis on the service sector, limited reliance on extractive industries, and stable macroeconomic fundamentals. Many wealthy and industrialised nations, particularly in Europe, fall into this group, reflecting their transition toward post-industrial economies with relatively sustainable environmental practices.

Cluster 2 – Pollution-burdened and Net-borrowing economies (n = 33) Countries in this group are marked by high particulate pollution damage, suggesting severe air quality challenges, likely due to urbanisation, traffic congestion, or biomass fuel use. At the same time, these economies demonstrate weak capital accumulation, as indicated by significantly negative gross savings and capital consumption figures. Natural resource depletion (e.g., carbon, energy, minerals) is moderate or low, and forest pressure is present but not extreme. These countries may be experiencing growth without sufficient reinvestment, making them vulnerable both environmentally and financially. Many African countries and small islands are part of this group.

Cluster 3 – Fossil-fuel-driven economies with strong reinvestment (n = 15) This cluster is defined by very high levels of carbon emissions and energy depletion, pointing to strong dependence on fossil fuels or energy-intensive industries. In contrast, particulate pollution and forest loss are relatively low, suggesting that environmental damage is concentrated in greenhouse gas emissions rather than direct land or air degradation. Economically, these countries show strong gross savings and capital reinvestment, indicating robust fiscal performance. Likely candidates include industrialised or oil-producing nations with high resource consumption but solid economic reinvestment strategies. Key members of this group are China, Saudi Arabia and Indonesia.

Cluster 4 – Extraction-intensive, resource-exporting economies (n = 4) Cluster 4 includes countries with extremely high mineral and forest depletion, pointing to economies heavily dependent on natural resource extraction. While carbon and particulate emissions are moderately elevated, they are not the dominant feature. Economically, these countries show high gross savings but low capital consumption, suggesting that they accumulate financial reserves from resource rents without heavily reinvesting in physical capital. These are likely mineral- or timber-exporting economies where environmental degradation is driven by global commodity demand rather than industrial activity or urbanisation. Key members of this group are Ghana and Guyana.

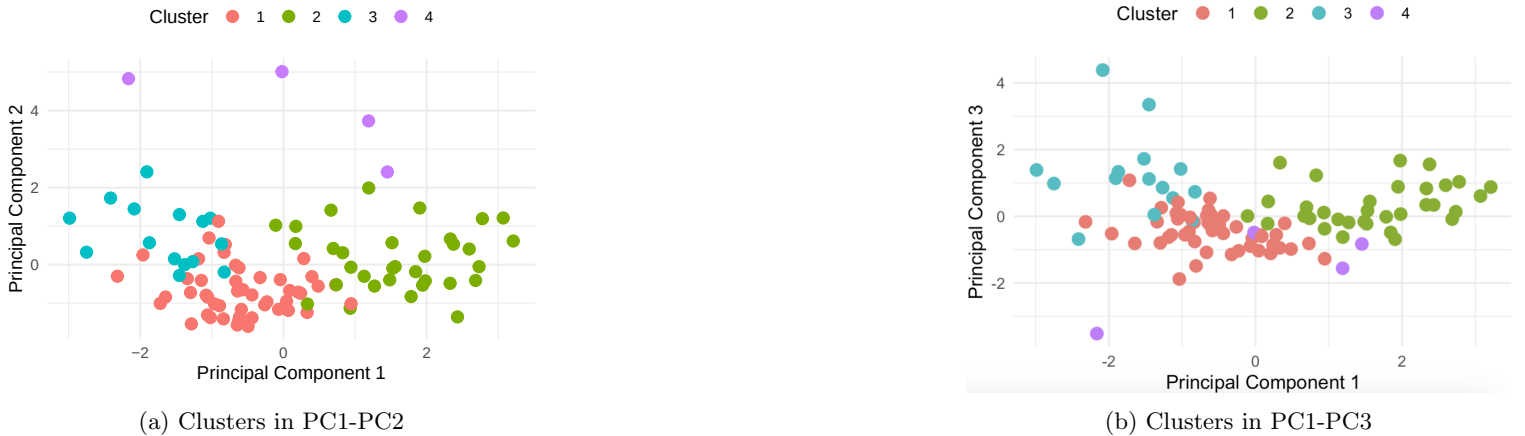


Figure 10: Clusters projected onto Principal Component planes

To conclude the second part of our analysis, we evaluate and briefly interpret how the clustering algorithm partitions the data along the principal components. The four clusters are clearly distinguishable along the first two PCs (Figure 10). There is a well-defined partition along the first component regarding the trade-off between local environmental degradation (e.g., particulate emissions, forest depletion) and capital-intensive growth strategies based on fossil fuel exploitation and reinvestment. There is also a more distinct partition along the second component, which reflects a gradient in cumulative environmental stress, across multiple ecological dimensions.

Although most of the clustering structure is defined by the first two components, a subtler but relevant division emerges along the third principal component (PC3). As Figure 10 shows, clusters 2 and 3 predominantly lie on the positive side of PC3, while clusters 1 and 4 tend toward the negative side. As previously discussed, PC3 contrasts countries with high energy depletion and capital consumption (positive values) against those focused on mineral extraction with limited domestic investment (negative values). This additional partition adds depth to our interpretation of cross-country sustainability profiles.

Cluster 1, composed of balanced, moderate economies, shows negative PC3 values, confirming their low-extraction, low-turnover profile. Cluster 4, characterized by extraction-intensive economies with limited reinvestment, also falls on the negative side, confirms their orientation toward mineral depletion and external resource flows with relatively low reinvestment in productive capacity. In contrast, Cluster 3, already characterized by high fossil fuel dependence with strong reinvestment, displays the highest and most dispersed PC3 values, reflecting internal heterogeneity but reaffirming its energy-intensive, capital-active profile. Finally, cluster 2 also lies on the positive side, though in a narrower range, suggesting modest capital use and energy draw. These countries may face environmental challenges but lack intensive investment strategies.

It is possible to observe graphically the distribution of countries among clusters in the related World Map in the Appendix.

4.3 Discriminant Analysis

This section aims to assess the robustness and interpretability of the clusters identified in the preceding cluster analysis by applying discriminant analysis techniques. Specifically, the chosen methods are Linear Discriminant Analysis (LDA), whose objective is to find a linear combination of features that best separates two or more classes in a dataset under the assumptions of homoskedasticity, and Quadratic Discriminant Analysis (QDA), which relaxes the assumption of equal covariance matrices for the classes. The ultimate goal is to compare these techniques, which will be used in the following two sections.

In order to implement both discrimination methods, it is essential to ensure that the variables are normally distributed. As pointed out in the first section of the project, we standardised and applied a logarithmic transformation to our variables to make them as close as possible to Normal distributions. Nevertheless, in order to interpret the results, we recognise that the variables are still not normally distributed, which will affect the results of the LDA and QDA.

Linear Discriminant Analysis (LDA)

To evaluate the predictive performance of the LDA, we employed the Leave-One-Out Cross-Validation (LOOCV) procedure. This method was chosen in light of the limited sample size - comprising only 100 observations - as it enables the most efficient use of available data. In LOOCV, the model is iteratively trained on $n - 1$ observations and tested on the single remaining observation, with this process repeated n times, where n denotes the total number of observations. This approach yields an almost unbiased estimate of the model’s generalisation error, while avoiding the loss of information associated with conventional data-splitting techniques. The use of LOOCV is particularly well-suited to small-sample contexts, where preserving the maximum amount of information for both model estimation and validation is essential. For the purpose of our analysis, the LDA models are estimated first on the original standardised and transformed data and subsequently on the principal components. This dual approach allows us to assess whether dimensionality reduction via PCA impacts classification performance and to determine which representation of the data provides more accurate and stable separation of the clusters.

Metric	Original Variables				Principal Components			
	Class 1	Class 2	Class 3	Class 4	Class 1	Class 2	Class 3	Class 4
Sensitivity	93.75%	90.91%	86.67%	100.00%	100.00%	87.88%	80.00%	75.00%
Specificity	90.38%	98.51%	98.82%	98.96%	86.54%	98.51%	100.00%	100.00%
Pos Pred Value	90.00%	96.77%	92.86%	80.00%	87.27%	96.67%	100.00%	100.00%
Neg Pred Value	94.00%	95.65%	97.67%	100.00%	100.00%	94.29%	96.59%	98.97%
Prevalence	48.00%	33.00%	15.00%	4.00%	48.00%	33.00%	15.00%	4.00%
Detection Rate	45.00%	30.00%	13.00%	4.00%	48.00%	29.00%	12.00%	3.00%
Detection Prevalence	50.00%	31.00%	14.00%	5.00%	55.00%	30.00%	12.00%	3.00%
Balanced Accuracy	92.07%	94.71%	92.75%	99.48%	93.27%	93.19%	90.00%	87.50%

Table 6: LDA classification metrics by class using original variables vs principal components (in percentages)

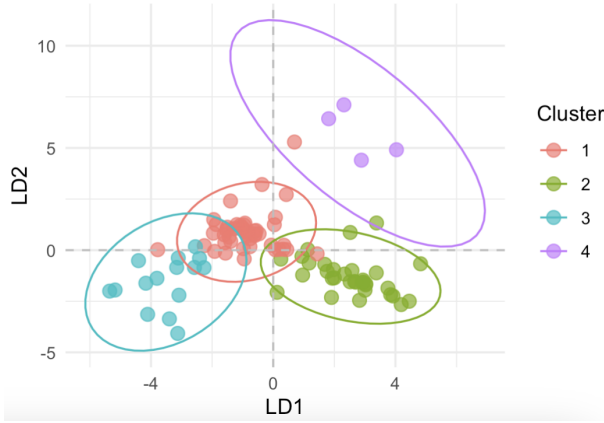
The LDA model applied to original variables exhibits strong classification performance, with Balanced Accuracy exceeding 92% across all clusters - indicating reliable prediction even under class imbalance. Sensitivity and specificity are consistently high, suggesting that the model effectively identifies both class members and non-members. Notably, Class 4 achieves perfect sensitivity (100%), though this class constitutes only 4% of the sample. This, along with a lower Positive Predictive Value (80%), signals potential overfitting: the model may be memorising rather than generalising. The Positive Predictive Value is high overall but lower for Class 4, reinforcing this concern. Detection prevalence also slightly exceeds the true prevalence for this minority class, indicating mild over-prediction. While Negative Predictive Value remains high across all groups, confirming low false negative rates, the results for small clusters warrant cautious interpretation. Overall, LDA performs robustly, but care is needed when evaluating performance on underrepresented classes.

The LDA model applied to principal components demonstrates consistently high performance across all classes, with Balanced Accuracy ranging from 87.5% to 93.3%. Sensitivity remains strong, particularly for Classes 1 and 2 (100% and 87.88% respectively), while still acceptable for Classes 3 (80%) and 4 (75%), despite their lower prevalence.

This indicates that the model maintains its ability to identify true class members even after dimensionality reduction. Specificity is excellent across all groups, reaching 100% for Classes 3 and 4, suggesting that misclassification of non-members is virtually absent for the minority classes. Precision (Positive Predictive Value) is perfect for Classes 3 and 4 and very high for Classes 1 and 2, indicating that the predicted labels are highly reliable. However, similar to the model based on original variables, Class 4 remains prone to potential overfitting, as indicated by its perfect precision and specificity despite low prevalence (4%). The detection prevalence closely matches the true prevalence for each class, suggesting no substantial bias in prediction frequency. Overall, LDA on the principal components achieves a level of predictive accuracy that is comparable to the model using original variables, but with slightly lower sensitivity for the minority class. This trade-off may reflect greater model stability with reduced risk of overfitting, due to dimensionality reduction.

Overall, both LDA models - whether based on the original transformed variables or on the principal components - exhibit strong classification performance and provide meaningful validation of the underlying cluster structure. The model estimated on the original variables shows slightly higher sensitivity for the smallest class but may risk overfitting due to perfect classification of low-prevalence groups. In contrast, the PCA-based LDA model achieves comparable overall accuracy and balanced performance, with slightly reduced sensitivity in minor classes, suggesting improved generalizability.

Figure 11: Clusters projected onto LD1–LD2 space



These results indicate that dimensionality reduction via PCA does not significantly compromise discriminative power and may help mitigate overfitting in small-sample, high-dimensional settings. It is also worth noting that Cluster 4 is distinctly characterized by variables such as mineral and forest depletion, which may naturally enhance its separability. The strong classification performance could therefore reflect genuine structural differences, rather than overfitting - though the small sample size still calls for caution.

Figure 11 displays the clusters obtained from LDA, projected onto the LD1–LD2 space. The plot based on the principal components is virtually identical and thus omitted for brevity.

From the visual inspection, it is evident that the cluster structure remains coherent and well-separated in the discriminant space, confirming that the robustness of the clusters is preserved even when transitioning from an unsupervised to a supervised classification framework. This consistency reinforces the validity of the clustering solution.

Quadratic Discriminant Analysis (QDA)

A fundamental assumption in LDA is the homogeneity among covariance matrices across groups, which will be relaxed in this section by using the QDA. The idea is to observe the consequences on robustness of using a more complex non-linear discriminant function. This added flexibility, however, comes at the cost of increased model complexity and a higher risk of overfitting - particularly problematic in small-sample settings. In fact, if we retain 4 clusters, then the fourth cluster's variance-covariance matrix results singular (non-invertible) due to the small number of observations and so the QDA fails. Therefore, due to the sensitivity of QDA to small group sizes, we reduce the number of clusters for this part of the analysis in order to ensure sufficient within-group observations and reliable covariance estimation.

	lsc_Carbon	lsc_Energy	lsc_Mineral	lsc_Forest	lsc_Part particulate	sc_Consume	sc_Gross
Cluster 1	-0.2847	-0.3686	-0.1688	-0.5322	-0.6867	0.2486	0.2027
Cluster 2	1.1750	1.3764	0.3437	-0.1430	-0.0777	0.3799	0.3975
Cluster 3	-0.4549	-0.4778	0.0010	0.9434	1.1354	-0.6807	-0.6223

Table 7: Three Cluster Characterisation

When reducing the number of clusters from four to three, we observe that Clusters 1 and 3 in the four-cluster solution remain largely stable, both in terms of their internal composition and their variable profiles. These groups are now captured by Clusters 1 and 2 in the three-cluster solution, and continue to reflect the previously defined categories of *Balanced*, *moderate economies* and *Fossil fuel-driven economies*, respectively. The main adjustment occurs with Cluster 2 of the original four-cluster solution, which is partially reabsorbed into the new Cluster 3, but retains its distinctive features - most notably high values in particulate emissions and low scores on capital-related variables - allowing it to still be interpreted as representing *Pollution-burdened and Net-borrowing economies*. This reallocation simplifies the cluster structure without significantly altering the substantive interpretation of the economic profiles.

Model	Original Variables	Principal Components
LDA Accuracy	96.00%	91.00%
QDA Accuracy	89.00%	89.00%

Table 8: LDA vs QDA Performance

The rationale for implementing QDA is to assess whether adopting a quadratic decision boundary leads to improved classification accuracy and more robust separation of the clusters. To this end, we compare the performance of LDA and QDA applied to both the original transformed variables and the principal components, under a three-cluster structure. As shown in the results, LDA consistently outperforms QDA in terms of overall accuracy across both data representations. An additional observation emerges when considering the redistribution of former outliers into the three clusters: while this adjustment leads to an improvement in LDA accuracy on the original variables, it is associated with a slight decline in performance when using principal components.

This analysis confirms the robustness of the cluster structure through discriminant methods. LDA outperforms QDA across all settings, particularly on the original variables, suggesting that the linear boundaries assumed by LDA align well with the data. Although QDA allows for greater flexibility, it does not improve performance - likely due to the small sample size and already well-separated clusters.

A key concern is the risk of overfitting, especially in LDA with the original variables, where some minority classes show near-perfect classification. In contrast, PCA appears to mitigate this effect slightly, producing more balanced results, though with a modest trade-off in accuracy. However, given the limited sample size, these findings should be interpreted with caution, as the models may not generalise reliably beyond the current dataset.

5 Conclusion

This report explored the sustainability profiles of 100 countries using PCA, Clustering, and Discriminant Analysis on seven environmental and economic indicators. PCA revealed three interpretable components summarizing trade-offs between local environmental degradation, capital-intensive fossil-fuel growth, and resource extraction. Clustering identified four distinct country groups, whose robustness was then confirmed through LDA and subsequently QDA.

The study provides a replicable framework for assessing sustainability in high-dimensional contexts. In particular, evidence from this dataset suggests that sustainable development strategies are shaped by differing balances between environmental burden and economic reinvestment capacity. Further analysis is necessary to test generalizability of our results, particularly in the discriminant analysis which suffered from limited sample sizes.

In fact, while the study’s strengths lie in its methodological coherence and interpretability, limitations include the cross-sectional nature of the data which dates to 1993 and reduced amount of observations. To this regard, a similar analysis on time-series data is suggested, to gain clearer intuition not only on what sustainable development strategies root, but also on how the factors which affect them evolve over time.

6 Appendix

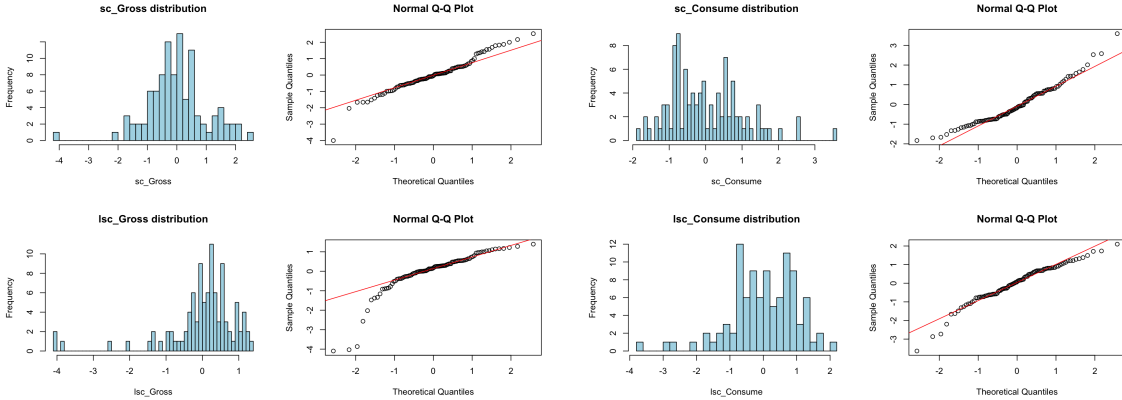


Figure 12: Gross distribution

Figure 13: Consume distribution

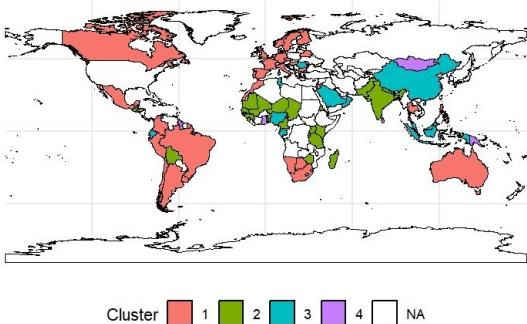


Figure 14: World Map by Country Clusters